

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – I
PAPER –2
TEST DATE: 20-11-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections: Section-A & Section-B**.

Section – A (01 – 04, 19 – 22, 37 – 40): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A (05 –10, 23 – 28, 41 – 46): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B (11 – 18, 29 – 36, 47 – 54): This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

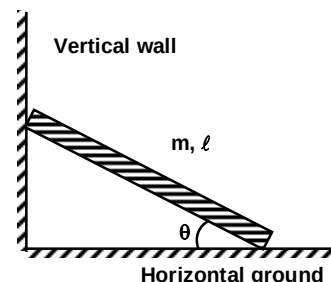
1. A uniform rod of mass m and length ℓ is released from the position shown in the diagram. If all the surfaces are smooth then find the Normal reaction applied by ground on the rod just after the release. $\left(\sin \theta = \frac{3}{5}\right)$.

(A) $\frac{9mg}{25}$

(B) $\frac{12mg}{25}$

(C) $\frac{16mg}{25}$

(D) $\frac{13mg}{25}$



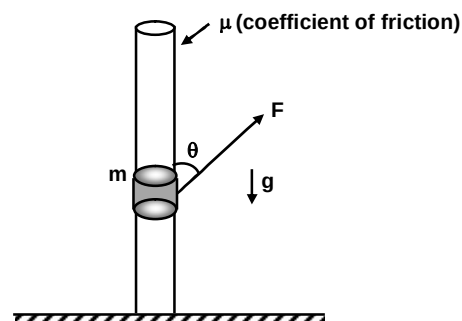
2. A collar of mass m is to be moved along a long fixed rough straight vertical rod by applying a force F which is constant in magnitude and whose direction is continuously changing such that θ is increased from 0° to 90° linearly with time. Initially the collar is at rest at ground and again comes to rest when the force becomes horizontal then

(A) $F = \frac{mg\pi}{2(1-\mu)}$

(B) $F = \frac{mg\pi}{(1-\mu)}$

(C) $F = \frac{3mg\pi}{2(1-\mu)}$

(D) $F = \frac{2mg\pi}{(1-\mu)}$



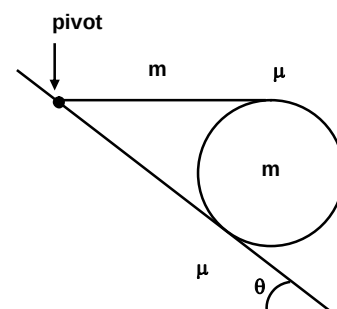
3. A horizontal stick of mass m has its left end attached to a pivot on the inclined plane while its right end rests on the top of a cylinder of same mass m as shown. The coefficient of friction between the cylinder and both the stick and plane is μ . Then the smallest value of μ for which the system doesn't slip anywhere is

(A) $\frac{\sin \theta}{2 + \cos \theta}$

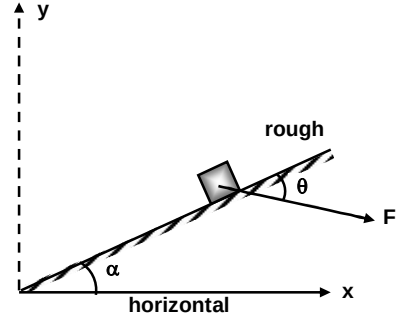
(B) $\frac{3 \sin \theta}{1 + \cos \theta}$

(C) $\frac{\sin \theta}{1 + \cos \theta}$

(D) $\frac{3 \sin \theta}{2 + \cos \theta}$



4. A block is placed on a fixed rough inclined plane ($\mu = \frac{1}{\sqrt{3}}$) as shown. A force F is applied on the block in the direction shown. Find the range of θ so that any value of the force F is not able to slide the block along the inclined plane. (Assuming that $\vec{F}$ always lies in x-y plane). (Also Given $\alpha < 30^\circ$).



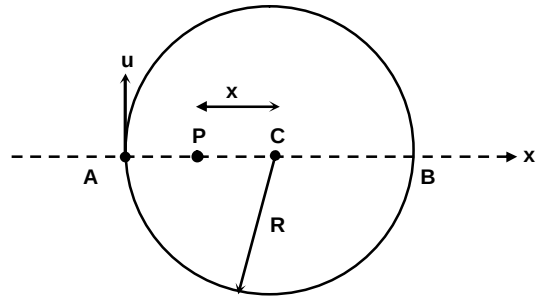
(A) $[60^\circ, 120^\circ]$
(C) $[30^\circ, 90^\circ]$

(B) $[30^\circ, 120^\circ]$
(D) $[60^\circ, 90^\circ]$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A bead of mass m is given a velocity u from point A along a fixed smooth circular wire of radius R which passes inside the bead as shown in gravity free space. A force F (constant in magnitude) always acting on bead towards a fixed point P. If u_0 is the minimum value of u so that bead completes the entire circle and if $\frac{u}{u_0} = \alpha$ and $\frac{x}{R} = \beta$ and $\vec{N}$ is the normal reaction experienced by bead at point B. Then choose the correct option(s).



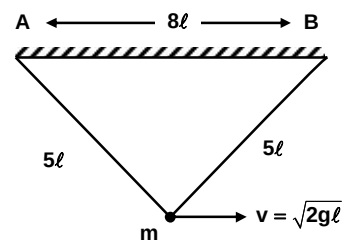
(A) If $\alpha = \sqrt{2}, \beta = \frac{1}{8}$ then $\vec{N} = \frac{F}{2} \hat{i}$

(B) If $\alpha = \sqrt{2}, \beta = \frac{1}{3}$ then $\vec{N} = -\frac{F}{3} \hat{i}$

(C) If $\alpha = \sqrt{3}, \beta = \frac{1}{8}$ then $\vec{N} = \frac{5F}{2} \hat{i}$

(D) If $\alpha = \sqrt{3}, \beta = \frac{1}{3}$ then $\vec{N} = \frac{-5F}{3} \hat{i}$

6. An ideal smooth massless inextensible string passes through the bead as shown whose ends are fixed at A and B. Initially the system is in equilibrium, now bead is given a velocity $\sqrt{2gl}$ in horizontal direction and in the plane of the string as shown. Let T_0 be the tension in the string and r_0 be the radius of curvature of particle just after the velocity is given. Then choose the correct option(s).



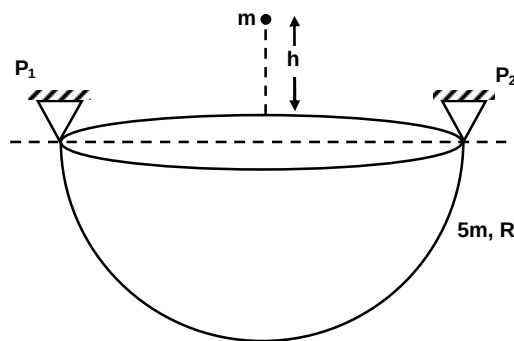
(A) $T_0 = \frac{31mg}{30}$

(B) $r_0 = \frac{35l}{3}$

(C) $T_0 = \frac{41mg}{40}$

(D) $r_0 = \frac{25l}{3}$

7. A small particle of mass m is dropped from height h from the flat surface of a uniform solid hemisphere of mass $5m$ and radius R which is held at rest with the help of two frictionless pivots on diametrically opposite points and the hemisphere is free to rotate about the axis passing through the pivots. The particle strikes on a point on circumference which is equidistant from the pivots and sticks to it. After the collision, the flat surface of hemisphere just able to reach in vertical plane. Then choose the correct option(s)



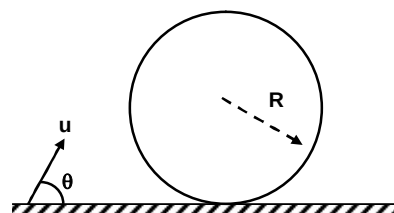
(A) $h = \frac{21R}{8}$

(B) $h = 3R$

(C) Angular velocity of the system just after collision is $\sqrt{\frac{7g}{12R}}$

(D) Angular velocity of the system just after collision is $\sqrt{\frac{8g}{12R}}$

8. A particle is thrown from ground with speed u at an angle θ with horizontal as shown. If u_0 is the minimum value of u so that the particle will cross the horizontal cylinder just grazing at two points on it and θ_0 is the corresponding angle of projection. Then choose the correct option(s).



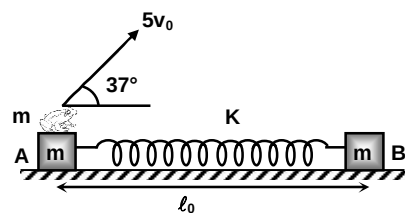
(A) $u_0^2 = Rg[2 + 2\sqrt{2}]$

(B) $\tan^2 \theta_0 = [3 + 2\sqrt{3}]$

(C) $u_0^2 = Rg[2 + 2\sqrt{3}]$

(D) $\tan^2 \theta_0 = [3 + 2\sqrt{2}]$

9. The system of two blocks and a frog (each having mass m) is placed on a smooth horizontal surface as shown. Initially the spring is in natural length (ℓ_0) and all are at rest. Now the frog jumps with velocity $5v_0$ at an angle of 37° with horizontal w.r.t. the block A and lands directly on block B. If R is the horizontal Range of frog in ground frame during the jump. If stiffness of the spring is



$K = \frac{\pi^2 mg^2}{288v_0^2}$, then choose the correct option(s).

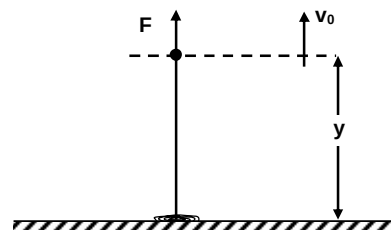
(A) $R = \frac{12v_0^2}{g}$

(B) $R = \frac{24v_0^2}{g}$

(C) $\ell_0 = \frac{12v_0^2}{g} \left[1 + \frac{2}{\pi} \right]$

(D) $\ell_0 = \frac{6v_0^2}{g} \left[3 - \frac{2}{\pi} \right]$

10. A uniform rope of linear mass density λ is coiled on a smooth horizontal table as shown. One end is pulled up with constant speed v_0 . Then choose the correct option(s)

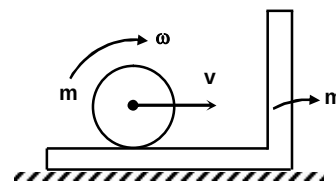


- (A) Force exerted (F) on the end of the rope is $\lambda y g + \frac{\lambda v_0^2}{2}$
 (B) Force exerted (F) on the end of the rope is $\lambda y g + \lambda v_0^2$
 (C) Heat loss per unit time is $\frac{\lambda v_0^3}{2}$
 (D) Heat loss per unit time is λv_0^3

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

11. An L shaped wedge of mass m having long horizontal base is placed on smooth horizontal surface as shown. A thin Ring of mass m and Radius R is set in motion on the rough horizontal part of the wedge with a linear velocity v and angular velocity ω such that $v = \omega R$. Ring collides with the smooth vertical part of wedge such that collision is perfectly elastic. After a sufficiently

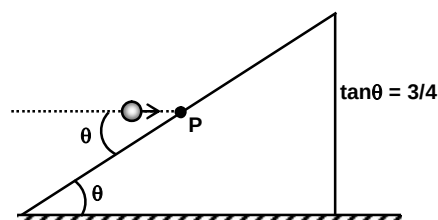


long time, the final velocity of wedge is $\frac{v}{K}$. Find the value of K .

(Assume wedge does not topple due to collision)

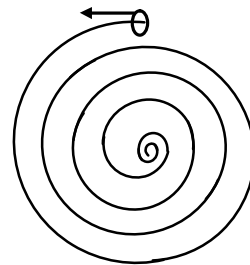
12. A projectile is thrown from ground with velocity 8 m/s. When it is moving with 6 m/s, it is at maximum separation from the point of projection corresponding to the speed 6 m/s. Then this maximum separation corresponding to speed 6 m/s is k meter. Find the value of k . (Take $g = 10 \text{ ms}^{-2}$)

13. A small elastic ball moving horizontally with 40 m/s collides with the smooth stationary wedge ($e = 1$) at point P which is placed on smooth horizontal surface as shown. After that it again strikes on the wedge at same point P after t_0 seconds. Find the value of t_0 . (Take $g = 10 \text{ ms}^{-2}$)

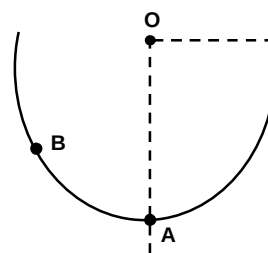


14. Bob of mass m of simple pendulum is given a velocity from the lower most position in horizontal direction in uniform vertical gravitational field (g) such that $\frac{T_{\max}}{T_{\min}} = 2$ during the subsequent motion. Then the tension in the string when it becomes horizontal is $K mg$. Find the value of K . (T represents tension in the string)

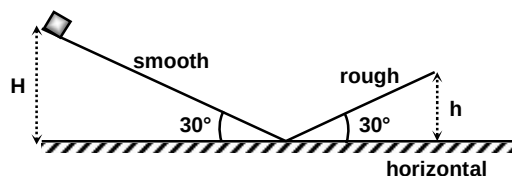
15. A bead of mass m is given a velocity along a smooth rigid spiral wire (mass $M = 2m$ and outer radius R) such that bead slides along the wire. When bead will reach the center of spiral and sticks to it, at that moment velocity of bead is v and angular velocity of spiral is ω . (Assume that the center of mass of loop lies at its centre and given that its moment of Inertia about an axis passing through centre and perpendicular to its plane is $I = \frac{MR^2}{2}$). Find $\frac{\omega R}{v}$ (Neglect gravity)



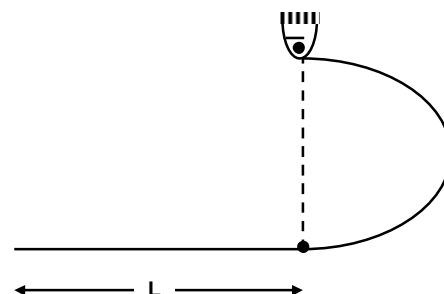
16. In a fixed hemispherical smooth bowl of Radius R , a particle is dropped at point B somewhere else other than point A. Collision with the bowl is perfectly elastic. For second collision to take place at point A, the maximum value of $\angle BOA$ is $\frac{\pi}{K}$ radian. Find the value of K .



17. A particle is released from the top of a fixed smooth inclined surface as shown from height H from the ground. The particle will reach upto height h from the ground on a rough fixed inclined plane which is directly connected to the smooth inclined plane. If coefficient of friction for rough inclined plane is $\frac{1}{2\sqrt{3}}$ the find $\frac{H}{4h}$. (Assume that particle does not jump.)



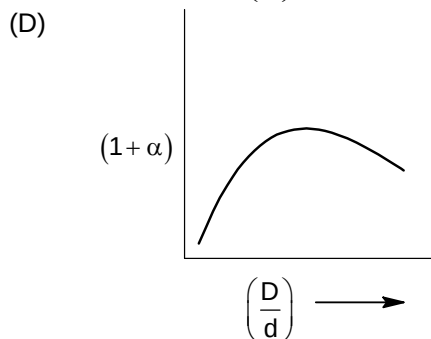
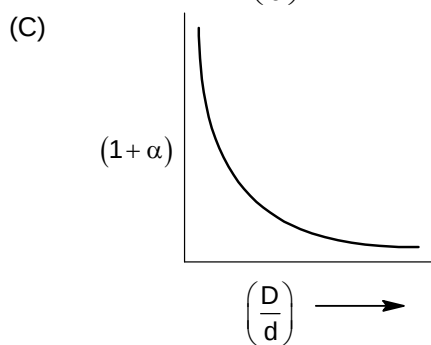
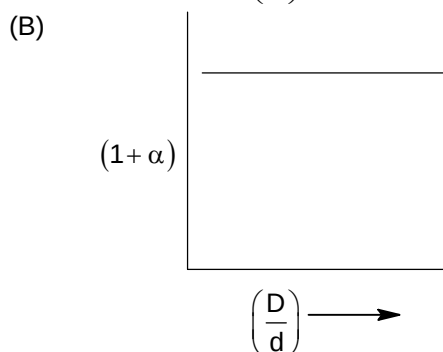
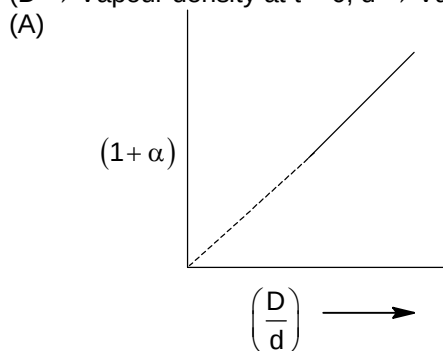
18. A uniform rod of length L is rigidly attached with a uniform semi-circular ring of same density as of rod to form a rigid body in a plane. The system is hanged in vertical plane in equilibrium with the help of hinge as shown such that rod is in horizontal position. Then the distance of centre of mass of the system from hinge is $\frac{L}{2} \left[\frac{\pi + \alpha}{\pi + \beta} \right]$. Find the value of $\alpha\beta$.



Chemistry**PART – II****Section – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

19. For the dissociation of PCl_5 (g) the variation of $(1 + \alpha)$ against (D/d) is represented as
 (D $\rightarrow$ Vapour density at $t = 0$, $d \rightarrow$ Vapour density at equilibrium)



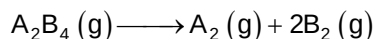
20. For an reaction $2 - \text{pentyne} \rightleftharpoons 1 - \text{pentyne}$. The temperature dependence of equilibrium constant is given by $\ln K = 8.0 - \frac{8000}{T}$. The value of ΔS° at 300 K. is....
- (A) $4R$
 (B) $8R$
 (C) $8000R$
 (D) $\frac{R}{8}$
21. An electron in an atom jumps in such a way that its kinetic energy changes from x to $\frac{x}{8}$. The change in its potential energy will be?
- (A) $+\frac{7}{8}x$
 (B) $+\frac{7}{4}x$
 (C) $-\frac{7}{4}x$
 (D) $\left(-\frac{9x}{4}\right)$
22. The time required for 50% completion of a first order reaction at 298 K is equal to that required for 75% completion of a reaction at 308 K. If the pre-exponential factor for the reaction is $3.5 \times 10^6 \text{ sec}^{-1}$ in the given temperature range. Find the ratio of K_{298} to K_{308} .
- (A) 0.5
 (B) 2
 (C) $\frac{3}{4}$
 (D) $\frac{3}{2}$

Section – A (Maximum Marks: 24)

This section contains SIX (06) questions. Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s) is (are) correct answer(s).

23. In the emission spectrum of H-atom from energy level 'n' to ground state in one or more step and no line belongs to the Breckett series is observed. The wave length of lines belonging to Balmer series may be (R = Rydberg constant)
- (A) $\frac{36}{5R}$
 (B) $\frac{16}{3R}$
 (C) $\frac{R}{3R}$
 (D) $\frac{35}{3R}$

24. For the reaction started with 1 mol $A_2B_4(g)$ is



The vapour densities are measured at various time of dissociation and observed value are

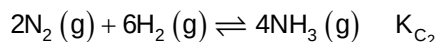
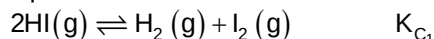
- | | | |
|-------|----------------------|---------|
| (i) | at time $t = 0$ min | VD = 50 |
| (ii) | at time $t = 10$ min | VD = 25 |
| (iii) | at time $t = 20$ min | VD = 20 |

If $\frac{V.D.(A_2B_4)}{V.D.(A_2)} = 5$ and $t_{100\%}$ is 40 min.

Choose the correct option(s)

- (A) The ratio of atomic mass of 'A' and atomic mass of 'B' is 1 : 2
 (B) Mole% of B_2 at time $t = 10$ min is 50%
 (C) Assuming uniform dissociation the ratio of rate of disappearance of A_2B_4 and rate of appearance of $A_2(g)$ between $t = 10$ min to $t = 20$ min is 1 : 1
 (D) Mole % of A_2 at $t = 10$ min is 50%

25. A closed rigid container of volume 1 litre, initially contains 4 mole of HI (g) and 4.25 moles of $N_2(g)$ at $27^\circ C$. When the mixture is heated to $127^\circ C$ two reversible reactions takes place till equilibrium is established.



At equilibrium the concentration of I_2 and NH_3 are 1.0 M and 0.5 M respectively. Choose the correct option

(A) $K_{C_1} = \frac{5}{32}$

(B) $K_{C_2} = \frac{16}{125}$

(C) $K_{C_1} \times \sqrt{K_{C_2}} = \frac{1}{25}$

(D) Conc. of N_2 at equilibrium = 4.0 mol/litre

26. How many of the following compounds has more than one $p\pi - d\pi$ bond

- (A) SO_2
 (B) SO_3
 (C) XeO_4
 (D) ClO_3^-

27. Which of the following statement(s) is/are true for $CH_2 = CH_2$ and $CH_2 = SF_4$ (Methyl sulphur tetrafluoride)?

- (A) In ethylene nodal plane lies in molecular plane
 (B) In methyl sulphur tetrafluoride nodal plane lies in axial plane
 (C) In methyl sulphur tetrafluoride nodal plane lies in equatorial plane
 (D) If the nodal plane lies in equatorial plane its stability will be less than if its lies in axial plane

28. How many of the following compounds has more than two $(3C - 2e)$ bond.

- (A) $(AlCl_3)_n$
 (B) B_2H_6
 (C) $Al(BH_4)_3$
 (D) $Be(BH_4)_2$

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

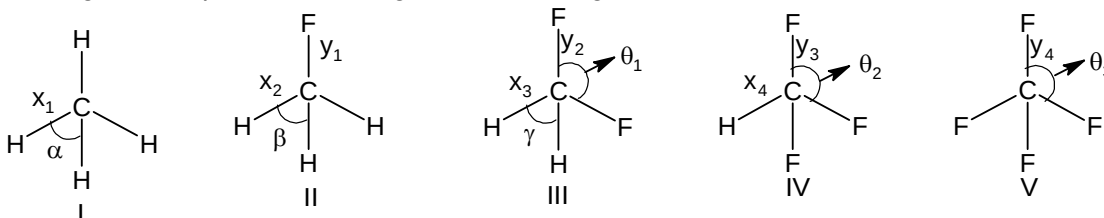
29. Not considering the electronic spin the degeneracy of the second excited state of H-atom is 'x' degeneracy of the second excited state of H^- is 'y'. And number of angular node in $d_{x^2-y^2}$ orbital is 'z'. Calculate the value of $\frac{x}{y} + z$?
30. The given reaction how many of them can give sp^2 hybridised boron centre. In case of the product(s)
- Borax $\xrightarrow{\Delta}$
 - Borax + $NH_4Cl \xrightarrow{\Delta}$
 - Borax $\xrightarrow{HCl}$
 - Borax $\xrightarrow{H_2O}$
 - $B_2H_6 + NH_3 \xrightarrow{\Delta}$
 - $B_3N_3H_3Cl_3 + LiBH_4 \xrightarrow{THF}$
 - $LiH + B_2H_6 \longrightarrow$
 - $K_3[AlF_6] + BF_3 \longrightarrow$
31. (i) If the sum of the number of lone pairs of electrons on each central atom in the given species is/are 'x'
 $[TeBr_6]^{2-}$, $[IF_2]^+$, SNF_3 and $[XeF_3]^-$
- (ii) and among the following species the number of diamagnetic spin is/are 'y'
 B_2 , Li_2 , C_2 , C_2^- , O_2^{2-} , O_2^+ , He_2^+ , Be_2 , KO_2 , K_2O , N_2O_4 , $B_3N_3H_6$
 (Atomic number of Te = 52, Br = 35, Xe = 54, K = 39, I = 127).
 Calculate the value of $\frac{x+y}{2}$?
32. Equilibrium constant of SO_3 (g) decomposition and formations was calculated by following two methods
- | | |
|---|--|
| <p>Method (1)</p> <div style="border: 1px solid black; padding: 10px; margin: 10px auto; width: fit-content;"> $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ </div> <p style="text-align: center;">I</p> <p>at $t = 0$, 10 moles of SO_3 is added 5(l) in container. If at equilibrium mixture is titrated with 0.2 M acidic $KMnO_4$ its volume 4 litre. K_C for the reaction is 'x' (mol/litre)
 The value of $\frac{y}{x}$ is?</p> | <p>Method (2)</p> <div style="border: 1px solid black; padding: 10px; margin: 10px auto; width: fit-content;"> $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ </div> <p style="text-align: center;">II</p> <p>Initially 2 : 1 molar ratio SO_2 and O_2 is added respectively. At the equilibrium total pressure of mixture 20 atm. Mole % of O_2 at equilibrium is 25%. K_P for the reaction is 'y' atm^{-1}.</p> |
|---|--|

33. Consider the following combination of atomic orbitals

Sr. No.	Combining orbitals	Internuclear axis
i.	$s + p_x$	(x)
ii.	$d_{xy} + d_{xy}$	(x)
iii.	$d_{yz} + p_z$	(z)
iv.	$p_y + p_y$	(y)
v.	$d_{yz} + d_{yz}$	(x)
vi.	$s + s$	(z)
vii.	$d_{z^2} + d_{z^2}$	(z)
viii.	$d_{xy} + d_{xy}$	(z)
ix.	$p_x + p_x$	(y)
x.	$s + p_z$	(x)

Calculate the value of $\frac{a^2 + b^2 + 2d}{4}$ (Where $a = \sigma$, M.O. $b = \pi$, M.O. $d = \text{non-bonding M.O.}$)

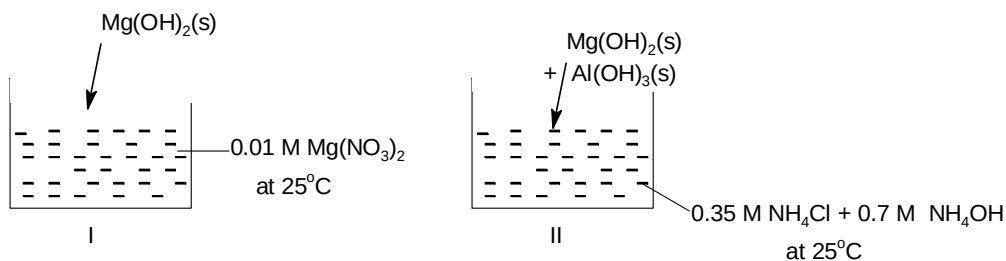
34. In the given compounds bond angle and bond length are defined



Order of bond parameter is/are given

- (I) $\gamma > \beta > \alpha$
 (II) $y_1 > y_2 > y_3 > y_4$
 (III) $\theta_3 > \theta_2 > \theta_1$
 (IV) $x_1 > x_2 > x_3 > x_4$
 (V) Dipole moment $\text{IV} > \text{II} > \text{III} > \text{I} = \text{V}$
 If the number of correct order(s) is 'x' and incorrect order (s) is 'y' about molecules I, II, III, IV and V. Calculate the value of $(2x + y)$ Is?

35. The solubility of
- $\text{Mg}(\text{OH})_2$
- in pure water is
- 5.8×10^{-4}
- g/litre at
- 25°C
- .



$$K_b \text{ for } \text{NH}_4\text{OH} = 1.0 \times 10^{-5}$$

$$K_{\text{SP}} \text{ Al}(\text{OH})_3 = 1.6 \times 10^{-20}$$

If the $[\text{Mg}^{+2}]$ in container (I) is 'x M' and concentration of $[\text{Mg}^{+2}]$ and $[\text{Al}^{3+}]$ in container II is 'y M' and 'z M' respectively. Calculate the value of $\frac{yz}{25x} = \dots ?$ ($\log 2 = 0.30$)

36. Choose the correct and incorrect statement in the following given statements
1. Hydrogen peroxide. In the pure state is non-planar and almost colourless.
 2. $S_8(s) + aOH^-(aq) \longrightarrow bS^{2-}(aq) + cS_2O_3^{2-}(aq) + dH_2O(l)$
The value of $a + b$ is 16.
 3. Hydrogen peroxide. In its reaction with KIO_4 and NH_2OH respectively is acting reducing agent and oxidizing agent.
 4. $MgCl_2 \cdot 6H_2O$ on heating give anhydrous $MgCl_2$.
 5. H_3BO_3 is mono basic weak Lewis acid.
 6. According to MOT B_2 and C_2 both do not have σ – bond.
 7. 'Be' cannot be qualitatively confirmed by the flame test.
 8. Correct order of density of alkali metals is $Li < K < Na < Rb < Cs$.
 9. The silicates in which three oxygen atoms of $[SiO_4]^{4-}$ are shared known as sheet silicate.
 10. Pb^{+4} compounds are stronger oxidizing agent than Sn^{+4} compounds.
 11. Graphite is better lubricant on the earth than on moon.
 12. The hydrolysis of trialkylchlorosilane R_3SiCl yields trialkylchlorosilanol.
- If the correct number of statement(s) is/are x and incorrect statement(s) is/are y . Value of $\frac{3x + y}{6}$ is....

Mathematics**PART – III****Section – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

37. If $f(x)$ is a function such that $f(x+y) + \frac{f(x) \cdot f(y)}{1+y^2} + \frac{\tan^{-1}x \cdot \tan^{-1}y}{1+y^2} = 0 \forall x, y \in \mathbb{R}$ and $f'(0) = 1$, then the function $f(x)$ is
 (A) $1 - \tan^{-1}x - 2e^{-\tan^{-1}x}$ (B) $-(1 + \tan^{-1}x)$
 (C) $e^{1-\tan^{-1}x}$ (D) $1 - \tan^{-1}x - e^{-\tan^{-1}x}$
38. A function $f(x)$ is such that the area enclosed by $|y| = f(x)$ between $x = 1$ and $x = a$ ($a > 1$) is given by $g(a) = a^4 + 6a^2 - 16a + 9$, then the area formed by the curve $k(x) = \frac{f(x)}{g(x)+5}$ between the lines $x = 2$ and $x = 3$ with the x -axis is
 (A) $\frac{1}{2} \ln\left(\frac{101}{22}\right)$ (B) $\frac{1}{2} \ln\left(\frac{131}{22}\right)$
 (C) $\frac{1}{2} \ln\left(\frac{141}{12}\right)$ (D) $\frac{1}{2} \ln\left(\frac{101}{12}\right)$
39. The value of $\int e^{x^2} (2x - \cot x)(x^2 - \ln(\sin x))(\operatorname{cosec} x) dx$ is equal to
 (A) $e^{x^2} (2x - \ln(\sin x) + 1)(\operatorname{cosec} x) + c$ (B) $e^{x^2} (2x - \ln(\sin x) - 1)(\operatorname{cosec} x) + c$
 (C) $e^{x^2} (x^2 - \ln(\sin x) + 1)(\operatorname{cosec} x) + c$ (D) $e^{x^2} (x^2 - \ln(\sin x) - 1)(\operatorname{cosec} x) + c$
40. Let C be a curve $f(x, y) = 0$ passing through $(1, 1)$ such that it is orthogonal to the family of circles $x^2 + y^2 - ax = 0$, where 'a' being parameter, then the shortest distance between the curve C and the parabola $x^2 = 2y + 1$ is
 (A) $\sqrt{2} + 1$ (B) $\sqrt{2} - 1$
 (C) $\sqrt{3} + 1$ (D) $\sqrt{3} - 1$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

41. Let $f(x) = (\cos^2 x) \cdot (e^{\tan x})$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then which of the following options is/are true?
 (A) $f'''(x) = 0$ has atleast one root in $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 (B) $y = f'(x)$ has a point of local maxima in $x \in \left(-\frac{\pi}{4}, 0\right)$
 (C) $f''(x) = 0$ has exactly one root in $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 (D) $\lim_{x \rightarrow \alpha} \frac{x^3 - 6x^2 + 11x - 6}{x^2 - 3x + 2} = -1$
 where ' α ' is the total number of points of maxima and minima of $y = f'(x)$

42. Let $f(x)$ be a function satisfying $f'(x) > 0$, $f''(x) > 0 \forall x \in (0, 1)$ and $f(0) = 0$, $f(1) = 1$, then which of the following options is/are true?
- (A) $f\left(\frac{1}{2}\right) \cdot f\left(\frac{1}{3}\right) < 1$ (B) $f^{-1}\left(\frac{1}{2}\right) \cdot f^{-1}\left(\frac{1}{3}\right) > 1$
- (C) $f\left(\frac{1}{3}\right) \cdot f^{-1}\left(\frac{1}{3}\right) < \frac{1}{3}$ (D) $f\left(\frac{1}{2}\right) \cdot f^{-1}\left(\frac{1}{2}\right) > \frac{1}{2}$
43. In a question, a student was given to find the derivative of the product of two functions 'f' and 'g'. The student by mistake thought $(fg)' = f' \cdot g'$ for his question $f(x) = \sqrt{\frac{x-1}{x-3}}$ and he got the correct answer. Given that $g(1) = e$. Which of the following options is/are correct?
- (A) $g(x)$ is discontinuous at one point
- (B) $\lim_{x \rightarrow \infty} g(x) = 1$
- (C) $\lim_{x \rightarrow \alpha} \frac{x^3 - 8}{x - 2} = 12$ where α is the number of points of discontinuity of $g(x)$
- (D) $g(x)$ is discontinuous at two points
44. Let $f(x) = \cos^{-1}\left\{\frac{1-x^2}{1+x^2}\right\}$ and $g(x) = \{f(x)\} \forall x \in \mathbb{R}$ (where $\{x\}$ denotes fractional part function of x), then which of the following options is/are true?
- (A) Total number of points where $f(x)$ is not continuous is 2
- (B) Total number of points where $g(x)$ is not continuous is 7
- (C) $f'(1) = f'(-1) = 1$ and $g(x)$ is not differentiable at $x = 0$
- (D) Total number of points where $f(x)$ is not continuous is 3
45. Let $f(x)$ be a function defined by $f(x) = \int_0^{3\pi/4} |x \cos t - 3 \sin t + \sqrt{2}x| \left(\frac{1 + \sqrt{2} \cos t}{(\cos t + \sqrt{2})^3} \right) dt$ where x is any real number, then which of the following options is/are true?
- (A) $f(x)$ is continuous and differentiable everywhere
- (B) $f(x)$ is continuous for all x but non-differentiable at two values of x
- (C) Minimum value of $f(x)$ is $\frac{3}{2}$ for $x \in \mathbb{R}$
- (D) Minimum value of $f(x)$ is $\frac{3}{4}$ for $x \in \mathbb{R}$
46. Let the function $f : (-\infty, \infty) \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ be given by $f(x) = \sin^{-1}\left(\log_3\left(\frac{x^2 - x + 1}{x^2 + x + 1}\right)\right)$, then which of the following options is/are true for $f(x)$?
- (A) $f(x)$ is injective function
- (B) $f(x)$ is surjective function
- (C) $f\left(\frac{1}{x}\right) = -f(-x)$
- (D) The equation $f(x) = x - 1$ has two distinct roots

Section – B (Maximum Marks: 24)

This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive**.

47. If $f(x)$ is a function satisfying $3f^2(x) - 4xf(x) + 2 = 0$ and the value of integral $\int \frac{(6x-2)f(x) - 4x^2}{(3f(x) - 2x)(x^2 - f(x))^2} dx$ is equal to $\frac{ax + bf(x)}{3x^2f(x) - 4x^3 + 2} + c$ (where a, b, c are constants), then the value of $(a + b)$ is
48. If the area bounded between the curves $|x| + y = 4$ and $\sqrt{|x|} + \sqrt{y} = 2$ is $\frac{16\lambda}{3}$ sq. units, then the value of λ is
49. The value of $\lim_{n \rightarrow \infty} \left(\lim_{m \rightarrow \infty} \left(\sum_{r=1}^n \left(\sum_{k=1}^{mr} \frac{mn^2}{(m^2n^2 + k^2)(n^2 + r^2)} \right) \right) \right)$ is $\frac{\lambda\pi^2}{160}$, then the value of λ is
50. Consider $f(x) = \frac{(x^2 - 1)(x^2 - 9)}{g(x)}$, $x \in \mathbb{R} - \{a, b\}$, where $g(x)$ is a polynomial of degree ≤ 4 . $f(x)$ has a removable type of discontinuity at $x = a$ and $x = b$. The following three informations are also known
 (i) $\lim_{x \rightarrow \pm\infty} f(x) = \infty$ (ii) $\lim_{x \rightarrow 1} f(x) = -4$
 (iii) $\lim_{x \rightarrow -3} f(x) = a$ non-zero finite number
 Let L and M denotes the number of points of discontinuity and the number of points of non-differentiability of $y = |f(|x|)|$ respectively in $\mathbb{R}$ (the set of all real numbers), then the value of $(L + M)$ is
51. If $\int_0^{4\pi} \frac{dx}{2 + \sin 4x} = \lambda \int_0^{3\pi/8} \frac{dx}{7 + \cos\left(\frac{8x}{3}\right)}$, then the value of $\frac{3\lambda}{16}$ is
52. Let $f(x)$ be a polynomial of degree 6 such that $f(2) = 85$ and attains its local minimum value 4 at points $x = -1$, $x = 1$ and $x = 5$. The local maximum value of $f(x)$ lies at $x = \alpha$ and $x = \beta$.
 If $\lim_{x \rightarrow (\alpha+\beta)} \left(\frac{x^2 - 9}{x^2 - 5x + 6} \right) = \lambda$, then the value of $[\lambda]$ is (where $[\cdot]$ represents greatest integer function)
53. If the value of $\lim_{x \rightarrow 0} \left(\frac{1}{x^{2022}} \int_{x^{2023}}^{x^{2022}} \frac{(1+t)^{\frac{1}{2021}} - 1}{t} dt \right) = \frac{1}{\lambda}$, then the number of positive divisors of λ is
54. If the area bounded by the curves $x^2 = y$, $x^2 = -y$ and $y^2 = 4x - 3$ is $\frac{m}{n}$, where m and n are relatively prime positive integer, then $|m - n|$ is